

RelGate: Evidence-Grounded Readiness Gates for LLM-Reviewed Cloud Changes

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Abstract—Pre-deployment cloud-change review asks whether observability, alerting, rollout, rollback, ownership, service-impact, and validation evidence is sufficient. LLMs may support this reliability task, but unstructured answers can block adequately documented changes or assert unsupported readiness. RelGate audits a fixed change bundle through seven configurable gates: each PASS must cite a verbatim bundle span; absent support is MISSING_EVIDENCE. We evaluate 12 synthetic scenarios (9 unsafe and 3 READY controls), three models, and three modes—freeform, checklist, and RelGate—using one temperature-0 call per case/model/mode (108 calls). Under strict rescoring of the frozen outputs, all modes produced zero observed false-ready decisions on 27 unsafe observations per mode. Freeform blocked 9/9 READY controls, checklist blocked 1/9, and RelGate blocked 0/9. RelGate’s mean per-response unsupported-claim fraction was 0.056 versus 0.208 for checklist-only review, and its observed decision accuracy was 1.000. These controlled results suggest evidence-grounded review can improve auditability and decision behavior under the evaluated conditions; they demonstrate feasibility, not production-readiness accuracy, and require larger, realistic studies.

Index Terms—software reliability, LLM-assisted review, evidence grounding, cloud changes

I. MOTIVATION

Reliable release processes require explicit monitoring, rollout, rollback, ownership, and validation information [1], [2]. Our fully specified control included dashboard and alert links, a staged rollout, rollback triggers, ownership, SLO analysis, and test results; freeform reviews nevertheless blocked it across all three models. Checklist reviews sometimes marked a gate satisfied without literal support, a source-traceability concern distinct from broader LLM hallucination taxonomies [3]. RelGate treats this as evidence auditing rather than open-ended advice.

RelGate is complementary to retrieval-augmented generation, which retrieves non-parametric context [4], and Chain-of-Verification, which self-checks an earlier answer through verification questions [5]. Structured-generation methods constrain response form [6]; RelGate instead audits one fixed bundle and requires each positive readiness judgment to be traceable to that bundle.

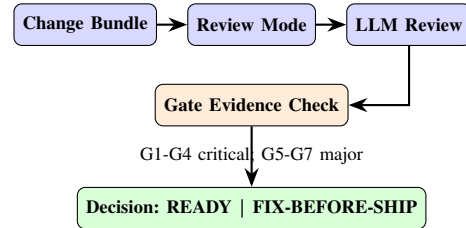


Fig. 1. RelGate workflow. The model must cite exact evidence for each passing gate; missing critical-gate evidence produces FIX-BEFORE-SHIP, while missing major-gate evidence raises a warning.

II. RELGATE DESIGN

The gates span detection (**G1** Observability, **G2** Alerting), containment/recovery (**G3** Rollout Safety, **G4** Rollback), accountability (**G5** Ownership), service-level consequence (**G6** Reliability Impact), and pre-release evidence (**G7** Validation). G1–G4 are blocking; missing evidence in G5–G7 raises a warning but does not by itself prevent READY. This is a pilot operationalization, not a universal taxonomy: organizations may change gate membership or criticality while retaining evidence grounding.

In this pilot, a PASS citation had to be a verbatim contiguous span from the supplied bundle after only case and whitespace normalization. The span could be shorter than its source sentence, but paraphrases, semantic equivalents, and prefix/suffix matches were unsupported; absent evidence required MISSING_EVIDENCE. Literal support does not establish relevance or operational adequacy, which remain separate evaluation dimensions.

III. PILOT STUDY

We constructed 12 gate-aligned scenarios: 9 with seeded gaps (e.g., absent alerts or rollback) and 3 fully specified READY controls. Each was reviewed once under **M0 Freeform**, **M1 Checklist** (G1–G7), and **M2 RelGate** (gates plus grounding) by GPT-5.5, Grok 4.3, and Llama 4 Maverick via OpenRouter at temperature 0: $12 \times 3 \times 3 = 108$ calls.

We report Critical Gap Recall, False-Ready Rate (unsafe approvals, $n=27/\text{mode}$), False-Block Rate (READY controls

This work was conducted independently and does not relate to the author’s employment. No employer resources, data, or confidential information were used.

TABLE I
CONTROLLED PILOT RESULTS ACROSS REVIEW MODES.

Metric	M0	M1	M2
Critical Gap Recall	0.975	1.000	1.000
False-Ready Rate ($n=27$)	0.000	0.000	0.000
False-Block Rate ($n=9$)	1.000	0.111	0.000
Unsupported Evid. Claims	0.035	0.208	0.056
Decision Accuracy ($n=36$)	0.750	0.972	1.000

blocked, $n=9/\text{mode}$), Decision Accuracy ($n=36/\text{mode}$), and Unsupported Evidence Claims. The latter is the mean of 36 response-level unsupported/total-claim fractions (zero if no claim). Scoring was deterministic: extracted quotes underwent the strict test above, and PASS on a seeded-missing gate also counted as unsupported. No human adjudication was used; model-level results and Wilson intervals are in the artifact.

IV. PRELIMINARY RESULTS

M0 found nearly all critical gaps (0.975) but blocked every READY control, yielding 0.750 accuracy. M0’s lower unsupported-claim fraction coincided with blocking all READY controls. M1 reduced false blocking to 0.111 and reached 0.972 accuracy, but its unsupported-claim fraction was 0.208. M2 produced zero observed false-ready and false-block decisions, 1.000 observed accuracy, and fewer unsupported claims than M1 (0.056). Thus, evidence grounding reduced observed false blocking while retaining zero observed false-ready approvals in this controlled pilot.

At model level, M2 was 12/12 correct for each model; its unsupported-claim fractions were 0.138 (Llama), 0.031 (GPT), and 0.000 (Grok), compared with 0.208, 0.083, and 0.333 under M1. In one M1 review of a queue-worker change with no alert evidence, Grok correctly blocked release but placed quotation marks around a paraphrased rollback statement, illustrating why decision correctness and quotation validity should be measured separately.

Uncertainty is substantial: for M2, 95% Wilson intervals are $[0, 0.125]$ for 0/27 false-ready and $[0, 0.299]$ for 0/9 false-block observations. These are zero observed errors, not zero risk.

V. DISCUSSION AND LIMITATIONS

This controlled, synthetic pilot demonstrates feasibility, not production effectiveness.

Because the scenarios and M2 policy share the same seven-gate ontology, the 1.000 decision accuracy may partly reflect instruction compliance under controlled conditions; it is not a production accuracy estimate. Clean READY controls omit evidence that is ambiguous, stale, conflicting, incomplete, or organization-specific. Each case/model/mode was observed once at temperature 0, so repeated-run robustness is unknown. Model versions and hosted implementations may also change; the named snapshots and frozen outputs bound these findings.

Some gates can be partially checked deterministically (e.g., presence of an alert definition or rollback field); RelGate studies the complementary evidence-to-decision problem over heterogeneous prose/configuration bundles, and a deterministic baseline remains future work. Future studies should use larger real/anonymized records, repeats, and expert/developer evaluation, separately measuring decision correctness, quotation validity, semantic relevance, and actual evidence adequacy. Such evaluation should also test whether evidence is current, mutually consistent, and sufficient under the organization’s own release policy rather than merely present.

Artifact: <https://github.com/lokesh0186/relgate>

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REFERENCES

- [1] B. Beyer, C. Jones, J. Petoff, and N. R. Murphy, Eds., *Site Reliability Engineering: How Google Runs Production Systems*. O’Reilly Media, 2016.
- [2] H. S. Gunawi, M. Hao, R. O. Suminto, A. Laksono, A. D. Satria, J. Adityatama, and K. J. Eliazar, “Why does the cloud stop computing? Lessons from hundreds of service outages,” in *Proc. 7th ACM Symp. Cloud Computing (SoCC)*, 2016, pp. 1–16.
- [3] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions,” *ACM Trans. Inf. Syst.*, vol. 43, no. 2, pp. 42:1–42:55, 2025.
- [4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.
- [5] S. Dhuliawala, M. Komeili, J. Xu, R. Raileanu, X. Li, A. Celikyilmaz, and J. Weston, “Chain-of-verification reduces hallucination in large language models,” in *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, 2024, pp. 3563–3578.
- [6] Z. R. Tam, C.-K. Wu, Y.-L. Tsai, C.-Y. Lin, H.-y. Lee, and Y.-N. Chen, “Let me speak freely? a study on the impact of format restrictions on large language model performance,” in *Proc. 2024 Conf. Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, 2024, pp. 1218–1236.